

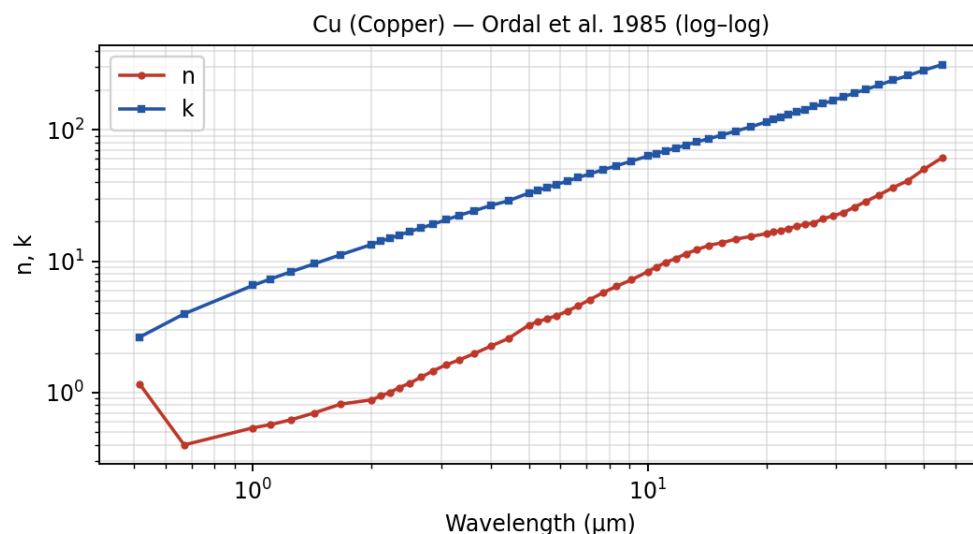
Optical constants of Cu (Copper)

Ordal et al. 1985: n, k 0.517–55.6 μm

Source: <https://refractiveindex.info/?shelf=main&book=Cu&page=Ordal> · Database record: database/data/main/Cu/nk/Ordal.yml (public domain, CC0 1.0)

주의: 적외선·원적외선 중심 데이터셋. 1 μm 아래로는 0.517 μm 와 0.670 μm 두 점뿐이어서, 그 사이(예: 532, 658 nm)를 보간해 쓰기에는 부적합함 — 구리는 이 구간에서 n, k가 급변함.

Complex refractive index ($n + ik$)



[재현 그림] 데이터베이스 레코드의 표 값으로 다시 그린 n, k 곡선 (원 페이지의 대화형 그래프에 해당, 로그-로그 축).

페이지 기본 표시값 ($\lambda = 0.517 \mu\text{m}$): $n = 1.1600$, $k = 2.6400$ · $\epsilon_1 = -5.6240$, $\epsilon_2 = 6.1248$ · $\alpha = 6.4169 \times 10^5 \text{ cm}^{-1}$ · $dn/d\lambda = -4.9739 \mu\text{m}^{-1}$ · group index $n_g = 3.7315$.

References

M. A. Ordal, R. J. Bell, R. W. Alexander, L. L. Long, M. R. Querry. Optical properties of fourteen metals in the infrared and far infrared: Al, Co, Cu, Au, Fe, Pb, Mo, Ni, Pd, Pt, Ag, Ti, V, and W. *Appl. Opt.* **24**, 4493–4499 (1985). <https://doi.org/10.1364/AO.24.004493>

Data (tabulated n, k)

λ (μm)	n	k	α (cm^{-1})	R	λ (μm)	n	k	α (cm^{-1})	R
5.17E-01	1.16E+00	2.64E+00	6.417e+05	0.6012	9.09E+00	7.21E+00	5.77E+01	7.977e+05	0.9915
6.70E-01	3.99E-01	3.97E+00	7.446e+05	0.9099	1.00E+01	8.31E+00	6.30E+01	7.917e+05	0.9918
1.00E+00	5.38E-01	6.53E+00	8.206e+05	0.9522	1.05E+01	9.00E+00	6.59E+01	7.887e+05	0.9919
1.11E+00	5.70E-01	7.32E+00	8.287e+05	0.9593	1.11E+01	9.77E+00	6.91E+01	7.823e+05	0.9920
1.25E+00	6.20E-01	8.31E+00	8.354e+05	0.9654	1.18E+01	1.05E+01	7.26E+01	7.732e+05	0.9922
1.43E+00	6.99E-01	9.53E+00	8.375e+05	0.9702	1.25E+01	1.14E+01	7.65E+01	7.691e+05	0.9924

1.67E+00	8.16E-01	1.12E+01	8.428e+05	0.9746	1.33E+01	1.22E+01	8.08E+01	7.634e+05	0.9927
2.00E+00	8.79E-01	1.34E+01	8.419e+05	0.9808	1.43E+01	1.32E+01	8.56E+01	7.522e+05	0.9930
2.11E+00	9.42E-01	1.42E+01	8.457e+05	0.9817	1.54E+01	1.38E+01	9.11E+01	7.434e+05	0.9935
2.22E+00	1.00E+00	1.49E+01	8.434e+05	0.9823	1.67E+01	1.47E+01	9.77E+01	7.352e+05	0.9940
2.35E+00	1.09E+00	1.58E+01	8.449e+05	0.9828	1.82E+01	1.54E+01	1.05E+02	7.250e+05	0.9945
2.50E+00	1.18E+00	1.68E+01	8.445e+05	0.9836	2.00E+01	1.62E+01	1.15E+02	7.226e+05	0.9952
2.67E+00	1.31E+00	1.79E+01	8.425e+05	0.9839	2.08E+01	1.68E+01	1.20E+02	7.250e+05	0.9954
2.86E+00	1.46E+00	1.91E+01	8.392e+05	0.9843	2.17E+01	1.70E+01	1.24E+02	7.181e+05	0.9957
3.08E+00	1.62E+00	2.06E+01	8.405e+05	0.9850	2.27E+01	1.77E+01	1.30E+02	7.197e+05	0.9959
3.33E+00	1.77E+00	2.23E+01	8.415e+05	0.9860	2.38E+01	1.84E+01	1.36E+02	7.181e+05	0.9961
3.64E+00	1.98E+00	2.42E+01	8.355e+05	0.9867	2.50E+01	1.91E+01	1.42E+02	7.138e+05	0.9963
4.00E+00	2.25E+00	2.65E+01	8.325e+05	0.9874	2.63E+01	1.95E+01	1.50E+02	7.167e+05	0.9966
4.44E+00	2.58E+00	2.88E+01	8.151e+05	0.9877	2.78E+01	2.10E+01	1.58E+02	7.142e+05	0.9967
5.00E+00	3.26E+00	3.30E+01	8.294e+05	0.9882	2.94E+01	2.21E+01	1.67E+02	7.138e+05	0.9969
5.26E+00	3.45E+00	3.45E+01	8.242e+05	0.9886	3.13E+01	2.34E+01	1.78E+02	7.146e+05	0.9971
5.56E+00	3.65E+00	3.63E+01	8.204e+05	0.9891	3.33E+01	2.57E+01	1.90E+02	7.170e+05	0.9972
5.88E+00	3.84E+00	3.83E+01	8.185e+05	0.9897	3.57E+01	2.85E+01	2.03E+02	7.146e+05	0.9973
6.25E+00	4.16E+00	4.07E+01	8.183e+05	0.9901	3.85E+01	3.19E+01	2.19E+02	7.148e+05	0.9974
6.67E+00	4.57E+00	4.33E+01	8.158e+05	0.9904	4.17E+01	3.62E+01	2.38E+02	7.172e+05	0.9975
7.14E+00	5.10E+00	4.62E+01	8.131e+05	0.9906	4.55E+01	4.09E+01	2.58E+02	7.126e+05	0.9976
7.69E+00	5.74E+00	4.95E+01	8.089e+05	0.9908	5.00E+01	5.00E+01	2.84E+02	7.138e+05	0.9976
8.33E+00	6.43E+00	5.32E+01	8.026e+05	0.9911	5.56E+01	6.12E+01	3.13E+02	7.074e+05	0.9976

λ , n , k 는 데이터베이스 레코드 원문 그대로 56행 (브라우저에서 계산한 해시와 대조해 전 행 일치 확인). $\alpha = 4\pi k/\lambda$ 와 $R = [(n-1)^2 + k^2] / [(n+1)^2 + k^2]$ (수직 입사, 벌크) 열은 원문에 없는 계산값.